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The future of live-streaming commerce: understanding the role of AI-powered virtual streamers

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Abstract

Purpose – The study aims to investigate how artificial intelligence (AI)-powered virtual streamers can supercharge brands in live-streaming virtual commerce (v-commerce). Built upon social identity theory (SIT) and experiential value theory, we developed a framework to investigate the impact of AI-powered virtual streamers' personalization and human-like personalities and live-streaming v-commerce's system quality and content quality on brand image, mediated by parasocial interaction and experiential value.

Design/methodology/approach – A survey was designed and distributed to the target respondents via social media channels. SmartPLS version 4.0.9.4 was used to analyze a total of 354 responses after the data were obtained via purposive sampling.

Findings – The results show that personalization, human-like personality, system quality and content quality are positively associated with parasocial interaction and experiential value, which subsequently impact brand image.

Originality/value – This study addresses the gap of relatively sparse academic literature on the implications of AI-powered virtual streamers in live-streaming v-commerce on brand image.

Keywords Live-streaming commerce, Virtual commerce, Parasocial interaction, Perceived value, Brand image, Consumer behavior

Paper type Research paper

1. Introduction

The potential of live-streaming e-commerce is bolstered with the gradual introduction of virtual reality (VR) technology and artificial intelligence (AI)-powered virtual streamers. Virtual streamers with AI capabilities are computer-generated personalities or characters who can mimic human-like interactions such as ascent, gestures and facial expressions. Retailers find using AI-powered virtual streamers efficient because they can work around the clock without any physical restrictions, lowering business operating expenses (Gao *et al.*, 2023). Additionally, AI-powered virtual streamers can improvise or alter their human-like characteristics based on brand attributes, enabling algorithms to resonate with customers in

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real-time (Hu and Ma, 2023). Moreover, brands find AI-powered virtual streamers appealing due to their lesser risk of scandal or potential threat of collaboration with competitors (Veloutsou and Mafe, 2020).

VR technology creates avenues for innovation advancement in retailing through virtual commerce (v-commerce). V-commerce offers customers a more authentic and engaging buying experience (Luna-Nevareza and McGovern, 2021) through virtual surroundings and immersion. In the combination of AI and VR-based technologies, AI-powered virtual streamers effectively work as salespeople or product demonstrators, while VR facilitates consumers to participate in immersive virtual showrooms, enabling 3D interaction with products, which is otherwise challenging. This combination leads to dynamic shopping experiences, resulting in increased sales and consumer satisfaction. This emerging business model is termed v-commerce, a subset of e-commerce that implements immersive technologies in retail commerce.

An example of such a combination is Alibaba's "Bye +" VR application, which enables consumers to shop in a 360-degree virtual world (Bye+, *n*.d.). Additionally, social media platforms are actively exploring the application of immersive live-streaming e-commerce to establish closer connections between brands and consumers. Live-streaming services with a combination of VR and AI-powered streamers by platforms such as Douyin (Chinese TikTok) and NTWRK (North American platform) combine VR with live-streaming, where AI-powered virtual streamers lead viewers and offer interactive features and tailored suggestions, demonstrating how these innovations may completely transform live-stream shopping (China Daily, 2024).

Brand image refers to the overall impression of a brand in the minds of consumers (Aaker *et al.*, 1995). Brand image is vital because it influences purchase decisions (Keller, 1993). The increasing popularity of live-streaming v-commerce poses challenges to retailers, such as dealing with high competition, the need for innovative brand differentiations and the frequent revision of value propositions to ensure their brand stands out. However, several unresolved issues remain, particularly concerning the impact of the new live-streaming technologies on brand image.

Firstly, live-streaming v-commerce platforms face challenges in establishing and maintaining authenticity and trust. Viewers may question the credibility and sincerity of AI-powered virtual streamers, product endorsements and promotional activities, particularly due to the lack of established rules and regulations (Chong *et al.*, 2023). Secondly, in recent years, the growth rate of the live-streaming platform user base has slowed down, and the demographic dividend in the live-streaming industry is gradually diminishing. Content and experience will be the key drivers for further profit growth. Yet, there is a lack of understanding in the live-streaming v-commerce context, thereby promoting the need to examine system quality and content quality and how they foster experiential value (Bai *et al.*, 2024). Thirdly, the study examines the role of parasocial interaction, as it has been proven essential in conventional live-streaming commerce (Ma *et al.*, 2023) but receives scant attention in live-streaming v-commerce in which virtual streamers are featured. This examination is crucial as consumers may perceive both streamers differently, leading to varying responses. Thus, the goal of this research is to investigate how brand image is affected when these cutting-edge technologies (AI-powered virtual streamers and VR environments) are integrated into live-streaming commerce.

2. Literature review

2.1 Review of live-streaming v-commerce literature

V-commerce is referred to as the "virtual channel" of retailing (Kaplan and Haenlein, 2009) or a "genre" of retailing that operates in the virtual channel (Rouse, 2011). This includes virtual stores for v-commerce, web stores for e-commerce and brick-and-mortar stores for physical

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commerce (Hofman-Kohlmeyer, 2017). As e-commerce recorded exponential growth post-2020, research on the integration of immersive technologies in e-commerce, as well as their adoption and acceptance, has gained momentum (e.g. Luna-Nevareza and McGovern, 2021). The widespread adoption of VR technologies has revolutionized both consumer experiences and retail strategies, giving rise to v-commerce, which is a subset of e-commerce that leverages virtual environments. This emerging form of economic exchange offers significant advantages, enhancing the purchase experience for consumers while providing retailers with innovative ways to engage their audience (Lombart *et al.*, 2020).

Researchers in the past have examined several v-commerce features and their influence on elements of customer journey stages (e.g. Martínez-Navarro *et al.*, 2019; Lombart *et al.*, 2020). Consumer attitudes toward v-commerce, along with its antecedents and consequences, were explored by Luna-Nevareza and McGovern (2021). Furthermore, online shopping in the context of virtual stores and virtual interactive shoppers was also investigated (Kang *et al.*, 2020). Previous literature also examines the impact of v-commerce elements on brand management and brand experience, particularly within the context of the metaverse (e.g. Wongkitrungrueng and Suprawan, 2023), where v-commerce is combined with social commerce. Pizzi *et al.* (2020) studied aspects of branding in the era of VR and highlighted the influence of VR elements on brand perception. As brands are relationship builders in the virtual world (Veloutsou and Mafe, 2020), further investigation is needed to understand the impact of v-commerce aspects on brand image.

Researchers are increasingly paying attention to the integration of live-streaming commerce platforms with VR coupled with AI-empowered virtual streamers (Bai *et al.*, 2024). Extant literature on live-streaming commerce has explored AI-empowered virtual streamers and their influence on customer satisfaction. Additionally, studies have investigated the impact of virtual streamers on purchase intention (Yao *et al.*, 2024), trust in virtual live streamers (Wang *et al.*, 2024) and the influence of anthropomorphism on consumers' willingness to accept virtual live streamers (Chen *et al.*, 2024) in the context of live-streaming commerce.

The existing studies also explore operations of AI-powered virtual streamers within evolving digital ecosystems like the metaverse (e.g. Allal-Chérif *et al.*, 2024). The broader aspects of consumer behavior, which are influenced by virtual streamers in the metaverse, include shopping behavior (Barta *et al.*, 2023), consumer arousal (Zhang *et al.*, 2023a, b) and trust (Wang *et al.*, 2024). However, studies investigating the impact of this joint innovation of VR integration along with AI-powered streamers on brand image remain unattended. The current development in VR-based retailing and the establishment of a VR marketplace make v-commerce and AI-powered streamers a vital topic for further investigation.

Please refer to Appendix 1 for key operational definitions.

2.2 Theoretical background and foundations

Burke (2006) identified social identity theory (SIT) as a social psychology explanation of how one's self-concept affects one's participation in groups and how one interacts with other groups. SIT offers a foundation from which psychologists and sociologists may comprehend the relationships and interactions that exist between people and the social environments in which they live (Liu and László, 2007). The idea focuses on how interpersonal and group settings differ based on the distinction between social and personal identity (Brown, 2000). The strength of the influence that interpersonal or intergroup interaction has on one's social identity is described by the interpersonal–intergroup continuum. The continuum shows that individuals differ in their susceptibility to influence, with some prioritizing interpersonal connections and others focusing on intergroup engagement. SIT aids in determining where an individual's socially produced self is located on this continuum. SIT suggests that

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interaction with a social referent is an important factor in the development of social identification, even if the interaction is unidirectional and one-to-many (i.e. parasocial interaction), which is demonstrated by the relationships between AI-powered virtual streamers and audiences).

In live-streaming v-commerce, AI-powered virtual streamers can be perceived by consumers as a representation of a social group or a unique identity of a virtual community. Viewers often identify with these virtual personalities, much as they do with human live-streamers who have cultivated significant influence (Ma *et al.*, 2023). Therefore, consumers may view an AI-powered virtual streamer as a member of their "in-group" and form a sense of social identification and belonging. This process of social identification can be influenced by the personalized and personified nature of the virtual streamer. AI-powered virtual streamers can process the data to understand the social identities of users (e.g. common interests and social trends), tailoring interactions to align with their group norms and values. Such provision of uniqueness or exclusivity in the content or any other form such as early access to products for members of specific groups can enhance their identification with the brand or platform.

For example, if the virtual streamer exhibits characteristics, behaviors or a persona that align with the user's own social identity or the identity of a group the user values, the user may be more likely to internalize the virtual streamer as part of their social identity. In terms of developing and deepening consumers' emotional connections with the brand or community, the immersive atmosphere of VR and empathetic language by AI-powered streamers could play a vital role. Past studies have shown that there can be an emotional connection between AI anchors and users (e.g. Zhang *et al.*, 2023a, b). Hence, we adopt SIT to explain that consumers identify with the personalization and human-like personality performance of AI-powered virtual streamers, which in turn affects their emotional (parasocial interaction) and perceptual responses (brand image). In live-streaming v-commerce, consumers may also form brand identification through the interaction with AI-powered virtual streamers and the immersive experience generated in VR live-streaming mode.

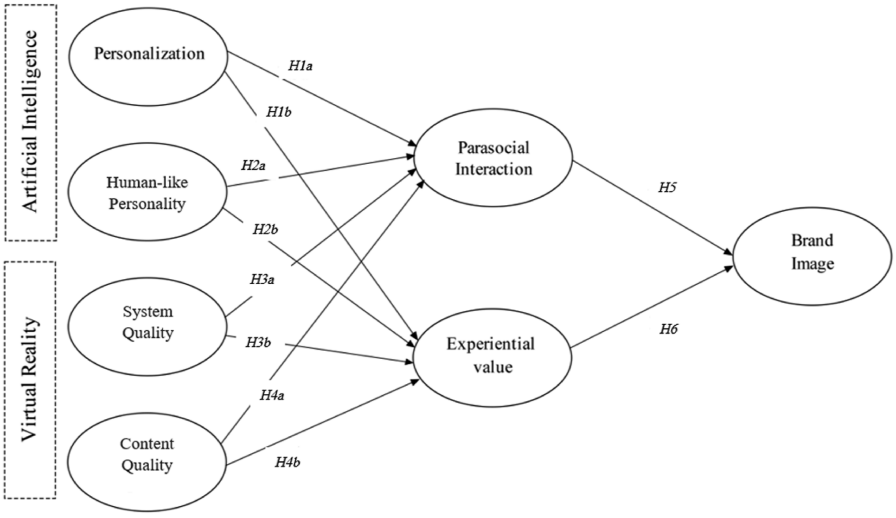
Next, the theory of experiential value suggests that the creation of value is not solely dependent on the intrinsic attributes of a product or service but also on the individual's subjective experience associated with it. The typology of experiential value put forward by Holbrook (1994) proposes a value landscape divided into four quadrants, with active and/or reactive value on one axis and intrinsic and/or extrinsic sources of value on the other. The quantity and nature of experiencing dimensions can vary (e.g. Brakus *et al.*, 2009; Dubé and Le Bel, 2003; Schmitt, 1999). It is important to note that these experience types can overlap, and multiple dimensions may be evoked simultaneously to create a holistic experience. Additionally, not all types of experiences may be equally important or relevant when considering technological impact.

This study focuses on the impact of AI and VR on experiential dimensions and posits that the personalization and human-like personality of AI-powered streamers, as well as features of v-commerce, can impact the overall experiential value. This is because these factors can influence strategic modules of "sense," "feel," "think," "act" and "relate." Specifically, we propose that these smart technologies have the potential to influence immersive experiences. If implemented and managed effectively, they can create new forms of experiential value. Against the aforementioned background, our study derives a conceptual model based on the theoretical underpinning of SIT and experiential value theory (see Figure 1).

2.3 Hypothesis development

2.3.1 Personalization and parasocial interaction. Personalization aims to create a more customized and engaging experience for users by adapting the AI-powered virtual streamer's content and delivery to suit their specific interests and requirements.

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Source(s): Authors' own

Figure 1.
Proposed conceptual framework

Personalization in AI-powered virtual streamers caters to the unique preferences and characteristics of individual viewers, which results in individual attention and relevance (Gao *et al.*, 2023). Personalized interactions make viewers feel valued and understood, leading to a stronger sense of connection and attachment to the AI-powered virtual streamer. According to past research, viewers' pleasant feelings and satisfaction are boosted by personalized interactions with AI-powered virtual streamers (Da Silva Oliveira and Chimenti, 2021). A stronger parasocial relationship is more likely to develop between viewers and the AI-powered virtual streamer when they believe that the AI-powered virtual streamer can satisfy their particular needs and interests.

H1a. Personalization of AI-powered virtual streamers positively affects parasocial interaction.

2.3.2 Personalization and experiential value. A sense of individual attention, relevance and satisfaction is produced when AI-powered virtual streamers are tailored to each user's specific tastes and attributes (Yao *et al.*, 2024). Consumers get a more engaging and rewarding experience because personalized interactions aid in delivering a sense of being valued and understood. Based on consumer preferences, previous behavior and demographics, personalized AI-powered virtual streamers can modify their content, recommendations and interactions (Hu and Ma, 2023). Consumers can feel more connected, engaged and fulfilled because of this customization, which improves experiential value (He, 2021; Zhang *et al.*, 2022).

H1b. Personalization of AI-powered virtual streamers positively affects experiential value.

2.3.3 Human-like personality and parasocial interaction. A unique and immersive experience can be created by AI-powered virtual streamers by hosting shows, delivering information and communicating with viewers (Chen *et al.*, 2024). According to Horton and Wohl (1956), parasocial interaction refers to viewers feeling a sense of connection, attachment and perceived relationship with media icons. Viewers frequently consider AI-powered virtual streamers as more relatable and likable when they are programmed to display human-like

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traits like facial expressions, gestures and conversational skills (Stein *et al.*, 2022). The process of giving virtual characters human-like traits and characteristics is known as personifying AI-powered virtual streamers (Chen *et al.*, 2024). The establishment of parasocial interaction is made easier when consumers perceive an AI-powered virtual streamer as a real person with unique qualities and attributes. Additionally, the human-like personality of AI-powered virtual streamers improves the consumers' sense of social presence (i.e. degree of awareness and experienced co-presence of others in a mediated setting) (Rice, 1993). Parasocial interaction is promoted, and the relationship between viewers and the AI-powered virtual streamer is strengthened by this increased social presence.

H2a. The human-like personality of AI-powered virtual streamers positively affects parasocial interaction.

2.3.4 Human-like personality and experiential value. In a variety of scenarios, including live-streaming e-commerce, the human-like personality of AI-powered virtual streamers intends to give consumers a more immersive and captivating experience (Zhang *et al.*, 2023a, b). According to past research, consumers find AI-powered virtual streamers to be more relatable, believable and interesting when they are made to act like real people (Chen *et al.*, 2024). Through the creation of an authentic social presence and emotional connection, the human-like personality of AI-powered virtual streamers improves the entire live-streaming commerce experience. In the context of AI technologies, it has been shown that anthropomorphic features can lead to the formation of optimal customer experience and service evaluation (Aw *et al.*, 2022; Wu *et al.*, 2023a, b). The human-like characteristics of AI-powered virtual streamers make the user experience more engaging and immersive. Users may experience increased happiness, fulfillment and entertainment while interacting with a personalized AI-powered virtual streamer, which will increase experiential value.

H2b. The human-like personality of AI-powered virtual streamers positively affects experiential value.

2.3.5 System quality and parasocial interaction. System quality refers to the quality of technical elements and features that consumers experience during their interactions with live-streaming v-commerce. This includes elements like visual fidelity, audio quality, motion smoothness, interaction and overall VR system performance (Lee *et al.*, 2020). Consumers' sense of presence and engagement with the content and media figure increases when they have a seamless and immersive VR experience (Liu *et al.*, 2024). Stronger feelings of connection and attachment are facilitated by system quality, including realistic visual and auditory elements, fluid interaction and the sensation of being physically present in the virtual environment (Liu *et al.*, 2024). This increased interest and intrigue may help users in the VR environment establish a stronger parasocial interaction with the AI-powered virtual streamers.

H3a. System quality in live-streaming v-commerce positively affects parasocial interaction.

2.3.6 System quality and experiential value. Consumers' sense of presence, engagement and enjoyment are increased when they have a seamless and immersive VR experience (Van Kerrebroeck *et al.*, 2017). A more immersive and fascinating experience is made possible by system quality elements, including realistic images, excellent audio, fluid interaction and a sense of being physically in the virtual environment. Additionally, consumers may experience a sense of novelty, excitement and intrigue due to the system quality aspects of live-streaming v-commerce. The total experience is improved, and consumers' attention is captured by VR's immersive qualities in combination with high system quality (Liu *et al.*,

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2024). Consumers' perceptions of the VR-enhanced environment as being more pleasurable, amusing and rewarding result in a perceived experiential value that is influenced by their increased engagement and curiosity.

H3b. System quality in live-streaming v-commerce positively affects experiential value.

2.3.7 Content quality and parasocial interaction. Content quality denotes the quality of the content that consumers are exposed to during their interactions in the VR-enhanced live-streaming environment (Liu *et al.*, 2024). This covers elements including the VR content's relevance, educational value, value as entertainment and general appeal. Grounded in interpersonal theories, it has been argued that communication (i.e. media content) determines the development of interpersonal relationships, and the same idea can be extended to the development of parasocial relationships (Aw and Chuah, 2021). According to past research, consumers' sense of connection and attachment to the media figure increases when they find VR content to be meaningful, educational, amusing and engaging (Van Kerrebroeck *et al.*, 2017).

H4a. The content quality in live-streaming v-commerce positively affects parasocial interaction.

2.3.8 Content quality and experiential value. Consumers' overall happiness, contentment and sense of fulfillment increase when they find VR content to be valuable, educational, amusing, emotionally engaging and appealing. The deep, compelling and meaningful experience made possible by the content quality ultimately raises the perceived experiential value. Consumers' overall enjoyment, engagement and contentment with the experience increase when they find VR content to be valuable and interesting (Jayawardena *et al.*, 2023; Liu *et al.*, 2024). Consumers' increased delight and fascination with VR content leads them to feel good and fulfilled, which increases the perceived experiential value.

H4b. The content quality in live-streaming v-commerce positively affects experiential value.

2.3.9 Parasocial interaction and brand image. Despite the physical distance, parasocial interaction can be seen in the communication and connection made between the streamer and consumers. Consumers develop a sense of familiarity, trust and attachment toward the streamer, which can positively influence their perception of the endorsed brand (Chinchilla and Kim, 2024; Hou *et al.*, 2024). It has been argued that the attachment individuals develop toward a celebrity, through their parasocial interactions, can lead to the transfer of both cognitive (credibility) and affective (attractiveness) perceptions from the celebrity to the brands, resulting in positive brand evaluation (Aw and Labrecque, 2020). Previous studies have examined the impact of parasocial interaction on various consumer behaviors, including brand perception and purchase intention (Lee and Watkins, 2016). When consumers engage in live-streaming v-commerce and develop a sense of parasocial interaction with the streamer, they are more likely to perceive the endorsed brand favorably. This improved brand perception, in turn, positively influences their intention to make purchases, as consumers tend to prefer brands they trust and feel connected to (Liu *et al.*, 2020).

H5. Parasocial interaction positively affects brand image.

2.3.10 Experiential value and brand image. Experiential value refers to the perceived benefits and enjoyment that consumers derive from their interactions with a product or service (Holbrook and Corfman, 1985). It encompasses sensory pleasure, emotional satisfaction and social interactions associated with a brand. In live-streaming e-commerce, where physical interaction with products is limited, experiential value plays a crucial role in shaping brand

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image. Studies have shown that when consumers perceive a product or service to provide a rich and enjoyable experience, they associate these positive experiences with the brand itself, resulting in enhanced brand image (Wu *et al.*, 2023a) or viceversa (Xie *et al.*, 2024). This association leads to increased consumer trust, loyalty and positive word-of-mouth (He, 2021).

H6. Experiential value positively affects brand image.

3. Methodology

3.1 Research design and data collection

A quantitative methodology with survey research design was employed to collect empirical data and analyze it using statistical methods. The target sample of this study was Chinese consumers aged 18 and above who have experience watching live-streaming v-commerce. Targeting those over the age of 18 was decided upon because Internet shopping has integrated itself into modern life, and this age group also includes those who have consistent sources of income, such as monthly salaries, living expenditures and scholarships. The survey questionnaires were disseminated via online channels, leveraging the absence of geographic limitations on the Internet to ensure the rapid acquisition of a large volume of data and to cover a more extensive consumer audience.

The present study employed the purposive and snowball sampling methods. In this study, the researchers started with the initial purposive sample and then leveraged the social connections of the initial respondents to locate additional respondents who also met the same criteria. Similar to past live-streaming studies (Chong *et al.*, 2023; Lo *et al.*, 2024), the sampling method aids in restricting the sample to live-streaming v-commerce consumers, given that the primary aim of the study involves understanding of live-streaming v-commerce experience. Only individuals who have experience viewing live-streaming v-commerce were provided access to the questionnaire to ensure the eligibility of possible responses. Although random sampling is robust, it was not employed for the current study considering the issue of sampling frame population diversity, and the requirement of platform-specific responses. It is challenging to obtain a sampling frame when it comes to identifying all consumers of live-streaming v-commerce platforms in a particular country (Kock, 2018).

According to the G-power calculation, the minimum sample size recommendation was 75. According to Kline (2017), any sample below 100 cases may not be recommended for any type of structural equation modeling (SEM) technique, unless the analysis is examining a very simple model. Furthermore, *a priori* inverse square root and gamma-exponential methods indicate 160 and 146 as the minimum sample size (Kock, 2018). Given this guidance, the initial target sample size of 75 may have been too small to reliably conduct the partial least squares structural equation modeling (PLS-SEM) analysis, which is a type of SEM technique. The current study involved 7 constructs and 21 items, resulting in a minimum number of 210 responses (Hair *et al.*, 2019). The substantially larger final sample size of 354 participants is more in line with the recommended guidelines and can be justified due to the complexity of the research model. As the study involved a complex model with multiple constructs and relationships, which would require a larger sample size to ensure stable and reliable estimates, the sample size of the current study is sufficient and justified. Table 1 shows the descriptive statistics of respondents' demographics.

3.2 Measurement instrument

The items for personalization were adapted from Zhang *et al.* (2022). The items for human-like personality were adapted from Chen *et al.* (2022a, b). The items for system quality and content quality for live-streaming v-commerce were adapted from Lee *et al.* (2020) and Yang

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Demographics	Item	Frequency	Percentage (%)
Gender	Female	180	50.85
	Male	174	49.15
Age range	18–25	105	29.66
	26–35	174	49.15
	36–45	70	19.77
	46 and above	5	1.41
Education level	High school	8	2.26
	Junior college	61	17.23
	Bachelor	206	58.19
	Master's	69	19.49
	Ph.D. and above	10	2.82
LSS viewing frequency	Daily	230	64.97
	Weekly	101	28.53
	Monthly	16	4.52
	Semiannually	7	1.98

Source(s): Analysis output

Table 1. Demographics of respondents ($n = 354$)

and Han (2021), respectively; the items for parasocial interaction were adapted from Liu *et al.* (2019), the items for experiential value were adapted from Mathwick *et al.* (2002) and the items for brand image were adapted from Aaker *et al.* (1995). The questionnaire items were revised or removed as needed based on the results from the pre-test and pilot-test to ensure their appropriateness and suitability for the study context (see appendix). Please find the final measurement items listed under Appendix 2 in the supplementary material file.

3.3 Data analysis plan

This study employed a three-stage analysis: data screening, measurement model validation and structural model assessment, as outlined by Hair *et al.* (2019). The data screening involved examining incomplete responses, straight-lining, outliers, normality and common method bias using IBM SPSS Statistics. The study utilized SEM to test the research model, which combines multiple regression and component analysis to evaluate theoretical models with both latent and observable variables. PLS-SEM was chosen for its advantages, including the ability to handle non-normal data and maximize the variance in dependent variables (Hair *et al.*, 2019). Additionally, importance-performance map analysis (IPMA) was used to determine the constructs with the most significant effects on the target construct (brand image) and their respective performance.

4. Analysis

4.1 Common method bias

To evaluate common method bias, we employed Harman's single-factor method (Babin *et al.*, 2016) (see Table 2). Harman's single-factor test results in a total variance of less than 50%, according to the unrotated principal components factor analysis. Additionally, the full collinearity test revealed (see Table 3) that the variance inflation factor (VIF) values were between 1.689 and 3.407. Since the highest VIF value was 3.407 (<10), multicollinearity is less likely a concern and further indicates fewer concerns of common method bias. A VIF value < 10 suggests that the predictor variable is in high correlation with other variables, leading to unreliable estimates of regression coefficients as well as inflated standard errors (Hair *et al.*, 2019).

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Component	Initial eigenvalues			Extraction sums of squared loadings		
	Total	% of variance	Cumulative %	Total	% of variance	Cumulative %
1	8.731	41.577	41.577	8.731	41.577	41.577
2	2.020	9.620	51.197	2.020	9.620	51.197
3	1.811	8.623	59.820	1.811	8.623	59.820
4	1.329	6.327	66.147	1.329	6.327	66.147
5	1.287	6.130	72.278	1.287	6.130	72.278
6	1.070	5.094	77.372	1.070	5.094	77.372
7	0.845	4.026	81.398			
8	0.513	2.442	83.840			
9	0.487	2.321	86.161			
10	0.374	1.780	87.941			
11	0.335	1.596	89.537			
12	0.308	1.467	91.003			
13	0.281	1.337	92.341			
14	0.264	1.259	93.600			
15	0.261	1.243	94.843			
16	0.227	1.079	95.922			
17	0.215	1.022	96.945			
18	0.199	0.949	97.893			
19	0.158	0.753	98.646			
20	0.154	0.732	99.379			
21	0.130	0.621	100.000			

Note(s): Extraction method: principal component analysis
Source(s): Analysis output

Table 2.
Total variance explained (CMB assessment)

Construct	Item	Outer loadings	VIF	Cronbach's alpha	CR	AVE
Brand image	BI1	0.905	2.521	0.890	0.932	0.820
	BI2	0.909	2.694			
	BI3	0.903	2.618			
Content quality	CQ1	0.889	2.278	0.875	0.923	0.800
	CQ2	0.906	2.620			
	CQ3	0.889	2.291			
Experiential value	EV1	0.921	2.870	0.852	0.911	0.773
	EV2	0.818	1.733			
	EV3	0.895	2.493			
Parasocial interaction	PSI1	0.931	3.407	0.904	0.940	0.839
	PSI2	0.884	2.418			
	PSI3	0.933	3.391			
Human-like personality	HP1	0.925	2.688	0.851	0.909	0.770
	HP2	0.873	2.080			
	HP3	0.833	1.943			
Personalization	PL1	0.876	1.689	0.827	0.893	0.737
	PL2	0.812	1.998			
	PL3	0.885	2.481			
System quality	SQ1	0.874	2.347	0.857	0.912	0.776
	SQ2	0.875	2.132			
	SQ3	0.893	2.033			

Source(s): Analysis output

Table 3.
Measurement model assessment

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4.2 Measurement model

To confirm the robustness of the measurement model, reliability and convergent validity were tested. Cronbach's alpha and composite reliability values exceed the necessary cutoff of 0.7 (Hair *et al.*, 2019) (see Table 3). All constructs show average variance extracted average variance extracted (AVE) values between 0.737 and 0.839, which are higher than the acceptable level of 0.5 (Hair *et al.*, 2019), and all item loadings are higher than the required minimum of 0.7. As a result, the findings showed that reliability and convergent validity are satisfied. To evaluate discriminant validity, we used the heterotrait-monotrait (HTMT) ratio, as recommended by Henseler *et al.* (2015). According to Table 4, the HTMT values varied between 0.230 and 0.701, all falling below the 0.85 cutoff. This serves as a confirmation of discriminant validity.

4.3 Structural model

We employed SmartPLS to compute T-values and path coefficients 5,000 times in order to bootstrap our way to a stable outcome (Hair *et al.*, 2019). According to Table 5, the findings demonstrated that personalization has a positive relationship with parasocial interaction ($\beta = 0.255$, $p < 0.001$) and experiential value ($\beta = 0.280$, $p < 0.001$). Human-like personality has a positive association with parasocial interaction ($\beta = 0.149$, $p < 0.010$) and experiential value ($\beta = 0.209$, $p < 0.001$). System quality shows a positive connection with parasocial interaction ($\beta = 0.275$, $p < 0.001$) and experiential value ($\beta = 0.229$, $p < 0.001$). The findings demonstrated that content quality has a positive correlation with parasocial interaction ($\beta = 0.320$, $p < 0.001$) and experiential value ($\beta = 0.278$, $p < 0.001$). There is a positive correlation between parasocial interaction ($\beta = 0.319$, $p < 0.001$) and brand image, as well as between experiential value and brand image ($\beta = 0.449$, $p < 0.001$). Based on the findings, all hypotheses were supported. With an R-square value of 0.451, the model accounts for 45.1% of the explained variance of brand image. The results of the mediating effects assessment test are presented in Table 5 as a post-hoc exploratory analysis. It is suggested that all mediating effects are significant.

4.4 Importance-performance map analysis

We further analyzed the data using an IPMA approach to glean more important information, as recommended by Ringle and Sarstedt (2016). This method uses the average latent variable scores to pinpoint a specific target construct and determine the importance of the most significant antecedent constructs. The IPMA diagram's variable representation for brand

	Brand image	Content quality	Experiential value	Parasocial interaction	Human-like personality	Personalization	System quality
Brand image							
Content quality	0.500						
Experiential value	0.701	0.610					
Parasocial interaction	0.610	0.628	0.581				
Human-like personality	0.510	0.434	0.519	0.463			
Personalization	0.456	0.322	0.595	0.539	0.351		
System quality	0.403	0.458	0.542	0.579	0.230	0.385	

Source(s): Analysis output

Table 4. Discriminant validity test using HTMT criterion

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image is shown in Figure 2. The Y axis displays the antecedent construct's performance, while the X axis highlights its importance in determining the target (consequence) construct. With a score of 0.455, experiential value is the most significant predictor, followed by parasocial interaction (0.344), with a significant difference in importance. Human-like

Hypothesis	Path	Std Beta	*T*-values	Decision
H1a	PL → PSI	0.255	6.502**	Supported
H1b	PL → VE	0.280	5.175**	Supported
H2a	HP → PSI	0.149	3.285*	Supported
H2b	HP → EV	0.209	3.847**	Supported
H3a	SQ → PSI	0.275	5.001**	Supported
H3b	SQ → EV	0.229	4.220**	Supported
H4a	CQ → PSI	0.320	6.136**	Supported
H4b	CQ → EV	0.278	5.608**	Supported
H5	PSI → BI	0.319	7.249**	Supported
H6	EV - > BI	0.449	10.146**	Supported
Post-hoc	SQ → EV → BI	0.103	3.953**	–
Post-hoc	CQ → PSI → BI	0.102	4.671**	–
Post-hoc	SQ → PSI → BI	0.088	3.982**	–
Post-hoc	HP → PSI → BI	0.048	2.860**	–
Post-hoc	HP → EV → BI	0.094	3.492**	–
Post-hoc	PL → PSI → BI	0.081	4.776**	–
Post-hoc	PL → EV → BI	0.126	4.667**	–
Post-hoc	CQ → EV → BI	0.125	5.089**	–

Table 5.
Hypotheses testing results

Note(s): *0.01, **0.001, PL = personalization, HP = human-like personality, SQ = system quality, CQ = content quality, PSI = parasocial interaction, EV = experiential value and BI = brand image
Source(s): Analysis output

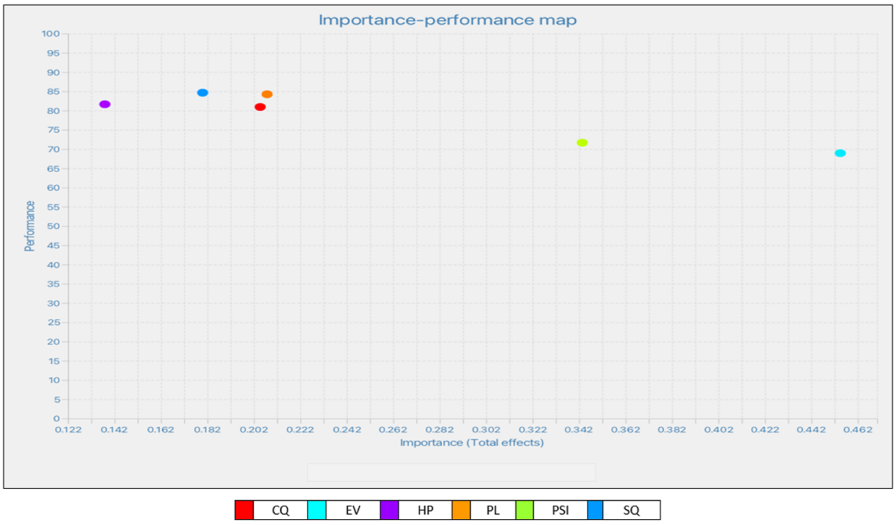


Figure 2.
IPMA results

Source(s): Analysis output

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personality with an IPMA performance value of 82 is the least significant predictor in terms of its influence, with a total effect of 0.138. Performance-wise, the other variables ranged from 65 to 85. With a value of 69, experiential value is the variable that performs the lowest.

5. Discussion and conclusion

The main purpose of our research was to investigate how the use of AI-powered virtual streamers influences brand image. We further looked into how consumers' parasocial interaction and experiential value affected brand perception in the setting of live-streaming v-commerce. Our research demonstrates that parasocial interaction and experiential value, which are strongly correlated with brand image, are significantly impacted by the AI-powered virtual streamers and selected live-streaming v-commerce attributes.

5.1 Theoretical implications

First, we integrated SIT and experiential value theory in our study to comprehend the effect of smart and immersive technologies (i.e. AI-powered virtual streamers and live-streaming v-commerce) on brand image. The study results indicated that parasocial interaction and experiential value have a positive effect on brand image. This aligns with the understanding that in the live-streaming v-commerce environment, when consumers develop parasocial interaction with AI-powered virtual streamers, they are more likely to develop favorable attitudes toward the brand (Liu *et al.*, 2020), thus influencing their perception and attitude toward the brand image.

The study stretches the role of parasocial interaction to the latest live-streaming v-commerce environment, suggesting a phenomenon where consumers can develop illusionary intimacy with non-human media personalities, and the one-way relationship can even bias their evaluation. Additionally, the study proves that the experiential value consumers derive from AI-powered virtual streamers and live-streaming v-commerce environments play a significant role in brand image. When consumers benefit from the commerce experience, they associate these positive experiences with the brand itself, thus forming a positive brand image. Therefore, our study validates the applicability of SIT and experiential value theory in the live-streaming v-commerce environment and provides a new theoretical perspective for brand image research.

Secondly, we also confirmed the impact of the personalization of AI-powered virtual streamers on parasocial interaction and experiential value. This study suggests that when AI-powered virtual streamers are personalized to cater to the unique preferences and characteristics of individual audiences, stronger parasocial interaction is formed between the audience and the AI-powered virtual streamers (Da Silva Oliveira and Chimenti, 2021). Consumers' sense of connection and involvement is increased by the personalization of AI-powered virtual streamers, which improves the experiential value. This aligns with past finding that indicates personalization fosters a sense of psychological comfort for customers, resulting in consumers' feeling that the service firm is attentive to their needs and cares about them. As a result, consumers develop an enhanced overall assessment of the value provided by the firm.

Thirdly, our study's findings show that personifying AI-powered virtual streamers has an effect on parasocial interaction and experiential value. In the context of live-streaming v-commerce, it was discovered that the human-like personality of AI-powered virtual streamers has a favorable and considerable impact on parasocial interaction and experiential value. According to this study, audience parasocial interaction tends to increase when AI-powered virtual streamers are created with more human-like qualities (Chen *et al.*, 2024). In general, consumers are able to obtain more enjoyable and immersive experiences with

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optimal human-like qualities of AI-powered virtual streamers (Yao *et al.*, 2024). According to Zhang *et al.* (2023a, b), human-like personality enhances the efficiency of the interaction between AI objects and the users, facilitating the service experience. Our findings contribute to the literature by evidencing that human-like appearance can be effective in the commerce setting, in which human-to-human interactions are valued during transactions as accustomed in traditional commerce settings.

Furthermore, the study confirms the impact of system quality in live-streaming v-commerce on parasocial interaction and experiential value (supporting H3a and H3b). The finding emphasized the important role of system quality in establishing parasocial interaction with media personalities in the VR environment and creating unique experiential value, which is seldomly explored, whereby past research typically focuses on information quality and interaction quality (Zhang *et al.*, 2023a, b). This finding extends the idea that the system environment in a v-commerce setting is vital not only for operation but also for relational marketing purposes. Additionally, our research results also indicate the impact of content quality in live-streaming v-commerce on parasocial interaction and experiential value. When users perceive live-streaming v-commerce content as valuable and engaging, they develop stronger parasocial interaction with media personalities in the VR environment and perceive higher experiential value (Jayawardena *et al.*, 2023).

5.2 Practical implications

In view of the relevance of personalization and human-like personality of AI-powered virtual streamers, live-streaming commerce retailers can accurately configure the personalized and humanized traits of AI-powered virtual streamers to align with brand image requirements, providing customers with authentic and emotional services and interactions. Retailers might want to leverage AI technology to feature virtual streamers that are highly realistic and personalized, particularly through imbuing human-like traits, behaviors and the ability to engage in genuine interactions. To note, despite the above-mentioned implications, practitioners are advised of significant challenges in implementing live-streaming v-commerce with AI-powered virtual streamers.

First, the high initial costs associated with the development of humanized virtual avatars powered by advanced natural language processing and real-time interaction capabilities in VR environments can be prohibitive for many businesses. Once developed, the deployment of such technology involves further complexities that require specialized expertise, which might be challenging to identify and hire. Moreover, there are potential issues related to data privacy and security, as live-streaming involves real-time data transmission, which must be protected against breaches. These challenges can hinder the widespread adoption and effectiveness of AI-powered virtual streamers in live-streaming v-commerce.

Apart from that, our research findings indicate that the application of VR technology in live-streaming v-commerce can positively influence brand image through parasocial interaction and experiential value. Retailers should pay attention to the system and content aspects of their live-streaming v-commerce. System-wise, they need to ensure low latency, high-definition video and reliable connectivity during the live-streaming session to create an immersive and enjoyable experience. In addition, retailers should consider diversifying and refreshing their content from time to time to prevent customer fatigue. Using different content formats, styles or even streamers can help keep their live-streaming sessions fresh and engaging. However, some challenges that might pose difficulty include the high requirement in terms of infrastructure (i.e. to handle increased traffic and concurrent users). Furthermore, maintaining consistent content quality can be challenging, as coordinating and aligning efforts across different stakeholders can add complexity to the process.

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The increased usage of AI-powered virtual streamers also poses several societal implications such as digital relationships and impacts on mental health. Our findings in the context of parasocial relationships warrant the identification of potential risks in this context. Furthermore, cultural shaping based on personalization and human-like personalities of AI-powered virtual streamers is another frontier of social impact handling in this context. Finally, the widely discussed issue around AI-powered virtual streamers includes the impact on employment as well as ethical considerations. Business and policymakers can consider drafting policies along these issues and ensuring social benefits of this technology.

5.3 Limitations and research prospects

Firstly, the study faces limitations in terms of generalizability, as it is based on single-country data and purposive and snowball sampling. Therefore, the model could be tested with live-streaming v-commerce consumers from different geographical regions as well as by adopting other probability sampling techniques to examine its predictive validity. Secondly, biases in measurement instruments may also pose a challenge to the generalizability of results. Future research could explore qualitative or mixed-method research to better capture the dynamics of consumer experience with live-streaming v-commerce. Thirdly, future researchers could include additional VR and AI-powered virtual streamer attributes in their investigations. Fourthly, the current study does not address aspects of ethics, privacy and transparency. Future researchers are encouraged to incorporate "dark side" factors when developing conceptual frameworks and empirical models to extend the current understanding of the field. Lastly, we suggest that future research consider the boundary conditions that might affect the proposed relationships (e.g. demographic and personal trait factors) to gain deeper insights.

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(The Appendix follows overleaf)

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Supplementary material

Appendix 1

Table A1. Explanation of key terms

Electronic commerce (E-commerce)	The purchase and sales of products or services online using Internet is called as e-commerce
Live-streaming e-commerce	Live-streaming e-commerce refers to the integration of live video streaming with e-commerce platforms, allowing sellers to showcase and sell products in real-time through online channels
Virtual commerce (v-commerce)	The use of virtual reality (VR) and other immersive technologies such as augmented reality (AR) or extended reality (XR) to create immersive shopping experiences. V-commerce can be considered a subset of e-commerce because it comprises buying and selling goods and services over the Internet, albeit through more immersive and interactive technologies
Virtual reality (VR)	VR is a technology that creates a simulated environment, allowing users to immerse themselves in and interact with a three-dimensional world. VR typically requires specialized hardware such as VR headsets, motion sensors and sometimes handheld controllers
AI-powered virtual hosts	AI-powered virtual hosts are digital entities or avatars that utilize artificial intelligence to interact with users in a human-like manner. These hosts can perform a variety of functions, ranging from customer service and personalized shopping assistance to engaging users in interactive experiences
Parasocial interaction (PSI)	The psychological relationship experienced by viewers in their encounters with performers or personas in the media
Experiential value (EV)	EV refers to the perceived value derived from the experiences associated with a product, service, or environment. It goes beyond the functional benefits and includes emotional, sensory, social, and intellectual aspects that enhance a user's overall experience
Personalization	Personalization refers to the creation of a more customized and engaging experience to suit their specific interests and requirements
Human-like personality (HP)	Human-like personality comprises characteristics that mimic human interactions, such as facial expressions, gestures, voice modulation and conversational skills
Content quality (CQ)	The quality of the content that users are exposed to during their interactions in the live-streaming v-commerce
System quality (SQ)	The SQ refers to the quality of technical elements and features of live-streaming v-commerce that consumers come across during their interactions
Brand image (BI)	The overall impression of a brand in the mind of current or potential consumers

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Appendix 2

Constructs	Items	Adapted from
Personalization	PL1: The AI-powered virtual streamers can help me establish my product needs PL2: The AI-powered virtual streamers can help me identify which product attributes best fit my needs PL3: The AI-powered virtual streamers can provide me with personal product customization based on my requirements	Zhang *et al.* (2022)
Human-like personality	HP1: The AI-powered virtual streamers have similar personality characteristics as humans HP2: I feel that the AI-powered virtual streamers have their own personality HP3: The AI-powered virtual streamer's speaking style is similar to the human being's	Chen *et al.* (2022a, b)
System quality	SQ1: The live-streaming v-commerce platform is easy to maneuver or navigate SQ2: The interface of the live-streaming v-commerce platform is user-friendly SQ3: The live-streaming v-commerce platform is interactive	Lee *et al.* (2020)
Content quality	CQ1: The live-streaming v-commerce platform provides up-to-date contents or information CQ2: The live-streaming v-commerce platform provides sufficient contents or information CQ3: The live-streaming v-commerce platform provides relevant content or information	Yang and Han (2021)
Parasocial interaction	PSI1: I look forward to watching the AI-powered virtual streamers PSI2: When I am watching the AI-powered virtual streamers, I feel as if I am part of their group PSI3: The AI-powered virtual streamers make me feel comfortable, as if I am with friends	Liu *et al.* (2019)
Experiential value	EV1: Shopping from the live-streaming v-commerce platform "gets me away from it all" EV2: Shopping from the live-streaming v-commerce platform makes me feel like I am in another world EV3: I get so involved when I shop from the live-streaming v-commerce platform that I forget everything else	Mathwick *et al.* (2002)
Brand image	BI1: The brand is up-to-date BI2: The brand is sincere BI3: The brand is charming	Aaker *et al.* (1995)

Table A2.
Measurement items

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Bin Xu obtained her MBA degree from Graduate Business School, UCSI University, Malaysia. Her research mainly focuses on the application of artificial intelligence in e-commerce live streaming, devoting herself to studying technology's impact on business and the development of marketing strategies. Currently, she operates an e-commerce service company in China, where she applies her professional knowledge into practice and continues to learn and explore in real-world scenarios.



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